

Typesetting Mathematics Part 2

1 Basics:

Superscripts are Like: A^2

Grouped Superscripts: A^{a+2}

Subscripts are Like: B_b

2 Nesting:

Nested Superscripts: $K^{n^{i+2}}$

Nested Subscripts: $M_{m_{l+2}}$

Combined Nesting: $C_{m_{l+2}}^{n^{i+2}}$

Messy Scripting: $D_{j_{K,V}^{2n}}^{x_i^2}$

3 Fractions:

Simple Inline Fraction: $(N+1)/3$

Complicated Fraction: $\frac{a^2+b^2}{a+b}$

$$\frac{\frac{a}{x-y} + \frac{b}{x+y}}{1 + \frac{a-b}{a+b}}$$

4 Roots:

Square Root: $\sqrt{16} = 4$

Cube Roots: $\sqrt[3]{8} = 2$

Compare $\sqrt[\mu]{M}$ to $\sqrt[\nu]{M}$

$$\frac{\sqrt[3]{-k+\sqrt{m^2+n_2}}}{(n+1)^2}$$

5 Text in Math-Mode:

TextnotSpacedwith : $\frac{2}{3}$ Compared to:
Normal Spaced Upright Text A_{Text}

6 General Expressions:

6.1 Case-based Functions:

$$F(x) = \begin{cases} Ae^{at}, & \text{if } x \neq 0 \\ 0, & \text{Otherwise.} \end{cases} \quad (1)$$

$$n! = \begin{cases} 1 & \text{if } n = 0 \\ (n-1)! \times n & \text{if } n > 0 \end{cases} \quad (2)$$

6.2 Function Evaluation and Delimiters:

Take a Look at this: $(e^{\frac{1}{2}})$ **Considering this** $(e^{\frac{1}{2}})$

Evaluation of a Function:

$$f\left(\frac{e^x}{\sin x}\right)\Big|_{x=0} = \inf \quad (3)$$

6.3 Equation-Lining:

Underlining an Expression:

$$A = P + \left(\frac{2}{n+1} + \underline{[2 \times m]} \right) \quad (4)$$

Over-lining an Expression:

$$A = P + \left(\frac{2}{n+1} + \overline{[2 \times m]} \right) \quad (5)$$

Over and Under-braced Expressions:

$$A = \overbrace{B+C+D}^{ANote} \times \underbrace{E+F+G}_{DNote} \quad (6)$$

6.4 Split Equations:

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = \lim_{x \rightarrow 0} \frac{e^x - 1}{x} = \lim_{x \rightarrow 0} \frac{\ln(1+x)}{x} = \lim_{n \rightarrow \infty} \sqrt[n]{n} = 1 \quad (7)$$

6.5 Side-aligned Equations:

$$\begin{array}{cc} \sqrt{a+b} \neq \sqrt{a}+b & \sqrt{a+b} \neq \sqrt{a}+b \\ \sqrt{a+b} \neq \sqrt{a}+b & \sqrt{a+b} \neq \sqrt{a}+b \end{array}$$

6.6 Stacking and Chemical Equations:

$$\begin{array}{c} x, \\ x_j \end{array}$$

And: $\begin{array}{c} x \\ x_j \end{array}$, is the same as: $\begin{array}{c} x \\ x_j \end{array}$

$$\sum_{i=0}^n$$

is the same as:

$$\sum_{i=0}^n$$

$$\sum_{\substack{j \text{ even} \\ j \rightarrow m \\ i \text{ odd} \\ i \rightarrow n}}$$

$$A \overset{to}{\rightarrow} B$$

$$2\text{Na} + \text{D}_2 \overset{\text{heat}}{\rightleftharpoons} 2\text{NaD}$$

$$\text{H}_2 + \text{D}_2 \overset{\text{Put Thing here}}{\longrightarrow} 2\text{HD}$$

$$\text{H}_2 + \text{D}_2 \overset{\text{Put Thing here}}{\longrightarrow} 2\text{HD}$$

$$\text{H}_2 + \text{D}_2 \overset{\text{Put Content}}{\underset{\text{Ni}}{\longrightarrow}} 2\text{HD}$$

$$\vec{K} = \overrightarrow{OR} + \overrightarrow{RB} = \overleftarrow{BO}$$

6.7 Integration and Differentiation:

Normal Diff: $f'(x) = 4x^{(4-1)} + \frac{d(x^2)}{dx} \cos(x^2) - \frac{d(\ln x)}{dx} e^x - \ln x \frac{d(e^x)}{dx} + 0$

Newton Notation: $\dot{x}, \ddot{x}, \ddot{\ddot{x}}$

Partial Differentiation: $\frac{\partial^2 x}{\partial t^2} = 0$

For Simple Integrals: $\int x \, dx$

For Definite Integrals:

$$\int_0^\infty x \, dx$$

6.8 Vectors and Matrices:

$$\begin{matrix} A & B \\ C & D \end{matrix}$$

$$\begin{pmatrix} A & B \\ C & D \end{pmatrix}$$

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix}$$

$$\begin{vmatrix} A & B \\ C & D \end{vmatrix}$$

$$\left\| \begin{matrix} A & B \\ C & D \end{matrix} \right\|$$

$$\vec{I} = \left\{ \begin{matrix} i_1 \\ i_2 \\ \vdots \\ i_n \end{matrix} \right\}$$

$$\begin{pmatrix} A_1 & A_2 & A_3 & \dots & A_n \\ B_1 & B_2 & B_3 & \dots & B_n \\ \dots\dots\dots\dots\dots & & & & \vdots \\ C_1 & C_2 & C_3 & \dots & C_n \end{pmatrix}$$

Tiny Inline Matrices are Like : $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$